Zhonghao (Jonathan) Shi

Contact Information

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Research Statement

My research focuses on developing and evaluating socially assistive robots and AI agents for applications in education and healthcare, ensuring that these systems are not only personalized to each user's unique preferences and needs but also aligned with the domain values and expectations of the broader communities in which they are deployed.

Research Interests

Human-Centered AI and Robotics, AI in education

Education

Ph.D., Computer Science, University of Southern California (USC)
– Research Advisor: Professor Maja J. Matarić *May 2020 - September 2025 (Expected)*

BSc, Computer Science, University of Southern California (USC)
– Research Advisor: Professor Maja J. Matarić
– GPA: 4.00/4.00 *Aug 2017 - May 2020*

BEng, Electronic and Electrical Engineering, University College London (UCL)
– Academic Advisor: Professor Martyn Fice
– GPA: 3.89/4.00 Sep 2015 - May 2017 (transferred)

Honors and Awards

AAAI/ACM SIGAI Innovative AI Education Award *Feburary 2025*
Hult Prize US National Champion - Global Finalist *July 2025*
Columbia Venture Competition Winner ($25,000) *April 2025*
USC Order of Arête *May 2025*
USC New Venture Seed Competition 1st Prize ($36,500) *May 2024*
USC Robotics George Bekey Service Award *May 2021*
USC Computer Science Award for Outstanding Research *May 2020*
USC Computer Science Outstanding Student Award *May 2020*
USC Provost Award (highest scholarship of transfer students) *May 2020*
USC Provost's Undergraduate Research Fellowship *Aug 2020*
USC Undergraduate Research Symposium First Award *Apr 2019*
UCL Goldsmid Prize for Outstanding Students (top 3 of the class) *May 2017*

CO-AUTHORED AWARDED RESEARCH GRANTS

NSF/NIH Smart Health and Biomedical Research in the Era of Artificial Intelligence and Advanced Data Science. PI: Maja Matarić, Co-PIs: Stefanos Nikolaidis, Bruna Martins-Klein. ($1,200,000 total) *Aug 2024 - Aug 2028*

NSF CISE Community Research Infrastructure (CCRI). PI: Maja Matarić, Co-PI: Mohammad Soleymani. ($100,000 total) *July 2022 - July 2023*

Work Experience

Sara Technology Inc.
Lead AI/ML Researcher & Engineer, Pasadena, CA *May 2024 - August 2025*
- Developing an AI-enabled speech therapy app to support and empower children with speech difficulties, their families and speech-language pathologists.

– Conducting child-centered AI/ML research to improve speech and language model performance and trustworthiness for child users.

USC Interaction Lab
Ph.D. Research Assistant, Los Angeles, CA *May 2020 - Present*
Advised by Professor Maja J. Matarić

- **Human-Computer/Human-Robot Interaction**: Developing an open-sourced socially assistive robot platform using the blossom robot with real-time multimodal sensing and socially interactive capabilities [github];
- **Human-Centered Machine Learning** Examined test-time domain adaptation methods to enable personalized child speech recognition [paper]; Applied supervised/unsupervised domain adaptation to design personalized supervised machine learning models outperforming non-personalized baselines for affective recognition to detect cognitive affective states for children with ASD [github, paper]; Applied supervised machine learning methods to model user engagement from children with ASD [paper]

J.P.Morgan Chase & Co.
Summer Research Associate, New York, NY *Jun 2023 - August 2023*
Mentored by Hsiang Hsu, Chun-Fu (Richard) Chen

- **AI Fairness** Worked on developing machine learning methods for fairness-aware data distillation for attribute classification. Our work on machine learning systems and methods for fairness-aware data distillation for attribute classification has been granted as a [US patent].

J.P.Morgan Chase & Co.
Summer Research Associate, New York, NY *Jun 2022 - August 2022*
Mentored by Chun-Fu (Richard) Chen

- **AI Trustworthiness** Worked on developing a sophisticated mechanism which gives data owners fine-grained control while retaining the maximal degree of data utility. More specifically we propose Multi-attribute Selective Suppression, or MaSS, a general framework for performing precisely targeted data surgery to simultaneously suppress any selected set of attributes while preserving the rest for downstream machine learning tasks [paper]. Our work on machine learning systems and methods for removal of attributes from multi-modality and multi-attribute data has been granted as a [US patent].

Publications:
* Indicates Co-first Author

Refereed Journal Articles

[1] **Shi, Z.**, Groechel, T. R., Jain, S., Chima, K., Rudovic, O., & Matarić, M. J. (2022). Toward personalized affect-aware socially assistive robot tutors for long-term interventions with children with autism. *ACM Transactions on Human-Robot Interaction (THRI)*, 11(4), 1-28.

[2] Jain, S., Thiagarajan, B., **Shi, Z.**, Clabaugh, C. and Matarić, M.J., 2020. Modeling engagement in long-term, in-home socially assistive robot interventions for children with autism spectrum disorders. *Science Robotics*, 5(39).

[3] Clabaugh, C., Mahajan, K., Jain, S., Pakkar, R., Becerra, D., **Shi, Z.**, Deng, E., Lee, R., Ragusa, G. and Matarić, M.J., 2019. Long-term personalization of an in-home socially assistive robot for children with autism spectrum disorders. *Frontiers in Robotics and AI*, 6, p.110.

Refereed Conference Papers

[4] Fu, M., **Shi, Z.**, Huang, M., Liu, S., Kian, M., Song, Y., & Matarić, M. J. (2025). Personalized Socially Assistive Robots With End-to-End Speech-Language Mod-

els For Well-Being Support. *In proceedings of the 2025 International Conference on Social Robotics (ICSR).*

[5] **Shi, Z.**, Srivastava, H., Shi, X., Narayanan, S., & Matarić, M. J. (2024). Examining Test-Time Adaptation for Personalized Child Speech Recognition. *In Proceedings of Interspeech 2025*

[6] **Shi, Z.**, Zhao, E., Dennler, N., Wang, J., Xu, X., Shrestha, K., Fu, M., Seita, D. and Matarić, M., 2025. HRIBench: Benchmarking Vision-Language Models for Real-Time Human Perception in Human-Robot Interaction. *In Proceedings of 2025 International Symposium on Experimental Robotics.*

[7] **Shi, Z.**, Chung, D.*, Du, Y., Zhang, J., Raina, S. & Matarić. Is AI Ready to Support Speech Therapy? A Content and Risk Analysis of AI-Enabled Mobile Apps for Speech Therapy. *In Proceedings of Interaction Design and Children (IDC) Conference 2025.*

[8] Dennler, N., **Shi, Z.**, Yoo, U, Nikolaidis, S., & Matarić, M. Modeling Personalized Difficulty of Rehabilitation Exercises Using Causal Trees. Submitted to The IEEE International Conference on Rehabilitation Robotics (ICORR) 2025.

[9] Dennler, N., **Shi, Z.**, Nikolaidis, S., & Matarić, M. (2024). Improving User Experience in Preference-Based Optimization of Reward Functions for Assistive Robots. International Symposium on Robotics Research (ISRR), Long Beach, CA, Dec 8-12, 2024.

[10] **Shi, Z.**, Landrum, E., O'Connell, A., Kian, M., Pinto-Alva, L., Shrestha, K., Zhu, X., and Matarić, M.J., 2024. "How Can Large Language Models Enable Better Socially Assistive Human-Robot Interaction: A Brief Survey" *In Proceedings of AAAI Spring Symposium on Impact of GenAI on Social and Individual Well-being.*

[11] **Shi, Z.***, O'Connell, A.*, Li, Z.*, Ayissi, J., Hoffman, G., Soleymani, M., and Matarić, M.J., 2024. "Build Your Own Robot Friend: An Open-Source Learning Module for Accessible and Engaging AI Education" *In Proceedings of the 2024 AAAI Conference on Artificial Intelligence, Feb 2024.*

[12] **Shi, Z.***, Chen, H.*, Velentza, A., Liu, S., Dennler, N., O'Connell, A., and Matarić, M.J., 2023. Evaluating and Personalizing User-Perceived Quality of Text-to-Speech Voices for Delivering Mindfulness Meditation with Different Physical Embodiments. *In Proceedings of 2023 ACM/IEEE International Conference on Human-Robot Interaction (HRI 2023), Mar-2023.*

[13] Zhou, E.*, **Shi, Z.***, Qiao, X., Matarić, M.J. and Bittner A.K., 2021. Designing a Socially Assistive Robot to Support Older Adults with Low Vision. *In proceedings of the 2021 International Conference on Social Robotics (ICSR).*

[14] Groechel, T., **Shi, Z.**, Pakkar, R. and Matarić, M.J., 2019, October. Using socially expressive mixed reality arms for enhancing low-expressivity robots. *In proceedings of the 2019 28th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)* (pp. 1-8). IEEE.

[15] Clabaugh, C., Jain, S., Thiagarajan, B., **Shi, Z.**, Mathur, L. and Mahajan, K., 2018. Attentiveness of children with diverse needs to a socially assistive robot in the home. *In proceedings of the 2018 International Symposium on Experimental Robotics.*

Refereed Workshop, Short Papers, or Preprint

[16] **Shi, Z.***, Shrestha, K., and Matarić, M.J., 2025. Can Large Language Model Agents Infer Human Actions and Motives in Strategic Decision Making? *Accepted in AAAI Conference on Artificial Intelligence (AAAI-25) Workshop on Advancing Artificial Intelligence through Theory of Mind, Feb-2025*

[17] **Shi, Z.***, O'Connell, A.*, and Matarić, M.J., 2021. Personalized Re-Engagement Feedback for Adaptive Socially Assistive Robot Interventions. *Robotics: Science and Systems (RSS-23) Workshop on Social Intelligence in Humans and Robots, Feb-2023.*

[18] Chen, C. F., Hu, S., **Shi, Z.**, Gulati, P., Moriarty, B., Pistoia, M., Piuri, V., & Samarati, P. (2022). MaSS: Multi-attribute Selective Suppression. arXiv preprint arXiv:2210.09904.

[19] Chen, H.*, **Shi, Z.***, Dennler, N., Humber, N. and Matarić, M.J., 2021. Your Voice of Mindfulness: Evaluating and Personalizing Text-to-Speech Voices for Mindfulness Practice *2022 IEEE International Conference on Robotics and Automation (ICRA 2022) Workshop on Sound for Robots.*

[20] **Shi, Z.** and Matarić, M.J., 2022. Recognizing Fine-Grained Cognitive-Affective Behaviors of Children with Autism Spectrum Disorder. *International Society for Research on Emotion conference.*

[21] **Shi, Z.**, Cao, M., Pei, S., Tarbox, J.J. and Matarić, M.J., 2021. Toward Personalized Automated Annotation of Targeted Behaviors of Children with ASD in Robot-Assisted ABA Interventions. *In proceedings of the 2021 Virtual Conference of Technology, Mind, and Behavior, APA.*

[22] **Shi, Z.**, Pei, S., Qiao X., Groechel T.R. and Matarić, M.J., 2021. Personalized Affect-Aware Socially Assistive Robot Tutors Aimed at Fostering Social Grit in Children with Autism. *ACM/IEEE International Conference on Human Robot Interaction (HRI) Workshop on Child-Robot Interaction and Child's Fundamental Rights.*

Patent

[23] **Shi, Z.**, Hsu, H., Chen, R., and Huang, W. (2025). Systems and methods for fairness-aware data distillation for attribute classification. U.S. Patent No.18/516,519. Washington, DC: U.S. Patent and Trademark Office.

[24] **Shi, Z.**, Chen, R., Hu, S., Moriarty, W., & Pistoia, M. (2024). Systems and methods for removal of attributes from multi-modality and multi-attribute data. U.S. Patent Application No.17/934,935. Washington, DC: U.S. Patent and Trademark Office.

Service

Conference/Symposium/Workshop Organizers

- Program committee member for the 15th Symposium on Educational Advances in Artificial Intelligence, collocated with AAAI-26
- Program committee member for the 40th Annual AAAI Conference on Artificial Intelligence
- Session chair for the 15th Symposium on Educational Advances in Artificial Intelligence, collocated with AAAI-25
- Lead organizer for 2025 AAAI Spring Symposium on Child-AI Interaction in the Era of Foundation Model
- Lead organizer for Robotics: Science and Systems (RSS-24) Workshop on Social Intelligence in Humans and Robots

Reviewer

- International Journal of Human-Computer Interaction
- International Journal of Child-Computer Interaction

– ACM/IEEE International Conference on Human Robot Interaction (HRI)
– International Conference on Robotics and Automation (ICRA)
– International Conference on Rehabilitation Robotics (ICORR)
– International Journal of Social Robotics
– Frontiers in Robotics and AI
– IEEE International Conference on Robot & Human Interactive Communication (RO-MAN)
– Oxford Intersections: AI in Society

STUDENT RESEARCH MENTORING

Students

– Misha Fu — Computer Science, USC Undergraduate Student
– Jingzhen Wang — Computer Science, USC Undergraduate Student
– Xinyang Xu — Computer Science, USC Undergraduate Student
– Sean Kim — Computer Science, USC Undergraduate Student
– Siqi Liu — Computer Science & Cognitive Science, USC Undergraduate Student
– Sophia Pei — Computational Biology, USC Undergraduate Student
– Lydia DiBlasio — Computer Science, USC Undergraduate Student
– Han Chen — Computer Science, USC Undergraduate Student
– Natalie Humber — Computer Science, USC Undergraduate Student
– Martin Liu — Computer Science, USC Undergraduate Student
– Flora Jia — Computer Science, USC Undergraduate Student
– Eunsook (Victoria) Shin — Computer Science, USC Undergraduate Student
– Maÿlis Whetsel — Computer Science, Columbia University Undergraduate Student
– Amanda Yao — Computer Science, UC Berkeley Undergraduate Student
– Riya Ranjan — SHINE program, High School Student
– Anishka Raina — SHINE program, High School Student
– Allen Wang — SHINE program, High School Student

OUTREACH

Disability Resource & Awareness Fair
University of Southern California, Los Angeles, CA — *Jan 2024*

Robotics Family Night
Monterey Hills Elementary, South Pasadena, CA — *May 2019, Nov 2019*

The Help Group STEM^3 Academy Visit
STEM^3 Academy, Los Angeles, CA — *Jun 2019*

USC Remote Robotics Open House
USC, Los Angeles, CA — *2020,2022,2023,2024*